

DNS FROM THE FIRST PRINCIPLES

ALEXY KHRABROV



FIRST PAIR PRESS

DNS from First Principles

Names, Packets, Authority, Recursion, and the Design of rbgdns

Alexy Khrabrov

2026-07-24

Contents

Preface	2
1 The problem DNS solves	3
1.1 Identity is not location	3
1.2 Requirements that pull in different directions	3
1.3 Roles, not just “DNS servers”	4
2 Names form a delegated tree	4
2.1 Labels and the root	4
2.2 Zones are administrative cuts	5
2.3 Wildcards are synthesis rules	5
3 Resource records: typed facts with lifetimes	5
3.1 The common envelope	5
3.2 Core types	6
3.3 Additional data is an optimization	6
4 Messages on the wire	7
4.1 The twelve-byte header	7
4.2 Name compression	7
4.3 Errors are protocol results	7
5 UDP, TCP, EDNS, and truncation	8
5.1 Why there are two transports	8
5.2 Truncation must preserve a valid message	8
6 Authority: answering from owned facts	9
6.1 Finding the closest relevant authority	9
6.2 tinydns data as a source language	9
6.3 CDB: compile once, read predictably	10
7 Recursion: discovering an answer	11
7.1 Iteration from the root	11
7.2 The cache is part of correctness	11
7.3 Forward zones and djbdns roots	12
8 DNSSEC: authenticating the chain	12
8.1 What ordinary DNS cannot prove	12
8.2 rbgdns validation policy	12
9 Zone transfer and secondary service	13

9.1	AXFR is a stream, not a giant datagram	13
10	The rbgdns program family	13
10.1	One suite, small purposes	13
10.2	Specialized responders	14
11	Client behavior and diagnostics	14
11.1	A query is more than sending bytes	14
11.2	Practical checks	15
12	Security engineering in rbgdns	15
12.1	The packet is hostile	15
12.2	Least privilege and filesystem boundaries	16
13	Time and logs: TAI64N	16
13.1	Why a DNS suite contains time tools	16
14	Running rbgdns under supervision	17
14.1	The service contract	17
14.2	Recommendations	17
14.3	A practical selection rule	19
15	Operating an authoritative service	20
15.1	Build, stage, verify, replace	20
15.2	Observe the right signals	20
16	Testing DNS software	20
16.1	Layers of evidence	20
16.2	Adversarial cases worth keeping	21
16.3	Conformance as an executable specification	21
16.4	Hardening found by conformance work	23
16.5	Benchmarks and evidence-driven optimization	23
17	Reading the rbgdns source	25
17.1	A path through the code	25
17.2	Design patterns to carry elsewhere	25
18	Where DNS ends	26
19	Appendix A: Configuration quick reference	26
20	Appendix B: Further reading	26

Preface

DNS is often introduced as “the Internet’s phone book.” That metaphor is useful for about a minute. A phone book is one database, published in editions, mapping people to telephone numbers. DNS is a distributed protocol for finding typed, time-limited statements in a delegated tree. It has many writers, many readers, caches between them, multiple transports, and rules for proving both presence and absence. Names can point to addresses, but they can also identify mail exchangers, service endpoints, authoritative servers, cryptographic keys, and arbitrary text.

This book develops DNS from those underlying problems. The first half builds a mental model independent of any implementation. The second half walks through rbgdns, a memory-safe Rust reimplementation of the djbdns suite. The aim is not only to explain what each program does,

but why its boundaries look the way they do: immutable compiled data for authority, a separate recursive cache, small diagnostic clients, foreground daemons, and stream-oriented logging.

The code is the final authority for `rgbdns` behavior. This book describes version 0.1.0 as built on 2026-07-23.

1 The problem DNS solves

1.1 Identity is not location

A network delivers packets to addresses. Humans and applications want stable identities. Those two things should not be fused.

Suppose a service is reached at 192.0.2.8. If that address is embedded in every configuration, moving the service requires changing every client. A name such as `api.example` introduces indirection:

`application` → `api.example` → 192.0.2.8 → packets

Indirection has a cost: another system must answer the middle question. Its benefit is that the service owner can change the answer without changing the application. DNS is the globally deployed mechanism for this indirection.

The mapping is not a function from one name to one address. One name may have several addresses. The answers may differ by client location. A mail domain may name several mail exchangers with preferences. A service may delegate a subtree to another organization. The useful abstraction is therefore:

(owner name, record type, class) → a set of resource records

The owner and type together select an RRset. “RRset” means all resource records with the same owner, type, and class. Implementations should normally treat the set as a unit because caches and DNSSEC signatures do.

1.2 Requirements that pull in different directions

A global naming system must satisfy conflicting demands:

- It must scale without one central database receiving every query.
- Different organizations must control different parts of the namespace.
- Changes must propagate, but cached answers are essential for performance.
- Replies should usually fit in one datagram, but some answers are large.
- Old implementations must coexist with protocol extensions.
- A client needs to distinguish “no such name” from “that name has no record of this type.”
- Operators need a way to transfer complete zones and to diagnose individual exchanges.

DNS answers these demands with hierarchy, delegation, caching lifetimes, compact binary messages, UDP plus TCP, explicit result codes, and typed records. Many operational surprises are direct consequences of those design choices rather than random quirks.

1.3 Roles, not just “DNS servers”

The phrase “DNS server” hides several jobs.

An **authoritative server** publishes data for zones it controls. It answers from configured facts and does not chase referrals on behalf of a client.

A **recursive resolver** accepts a question from a stub client, follows the delegation chain, validates and caches what it learns, and returns a final answer.

A **stub resolver** is the client-side library or program that sends a recursive query to a configured resolver.

A **forwarder** sends selected questions to another resolver rather than performing iteration itself.

Keeping these roles distinct is both conceptual hygiene and a security boundary. An authoritative daemon does not need a large mutable Internet-fed cache. A recursive resolver does not need the private machinery used to edit a zone. `rgbdns` follows the `djbdns` design and runs authority and recursion as different programs.

2 Names form a delegated tree

2.1 Labels and the root

A DNS name is a sequence of labels. In the presentation form `www.example.com.`, the dots separate the labels `www`, `example`, and `com`. The final dot represents the root’s empty label. Reading from right to left walks from general to specific:

```
.           root
└─ com.     top-level domain
   └─ example.com. delegated domain
      └─ www.example.com.
```

The absolute name has a wire limit of 255 octets, including length bytes and the terminating root label. Each ordinary label is at most 63 octets. DNS names are not inherently UTF-8 strings. Internationalized names are normally converted by applications into ASCII-compatible labels before DNS sees them.

DNS comparison is case-insensitive for ordinary ASCII letters, although the original spelling can be preserved. A robust implementation therefore needs a canonical comparison rule without losing the bytes required for encoding.

In `rgbdns`, `src/name.rs` represents a name as a vector of byte-vector labels. Construction validates label and total lengths. Parsing accepts the familiar dotted form and backslash escapes. The type provides parent and subdomain operations, case-insensitive ordering, display formatting, and wire encoding. Making invalid names difficult to construct removes repeated checks from the rest of the system.

2.2 Zones are administrative cuts

The namespace is one tree; a zone is an administratively served portion of that tree. The two are not identical.

The zone `example.com.` might contain records for `www.example.com.` and `mail.example.com.`, then delegate `research.example.com.` to other servers. The child remains below `example.com.` in the namespace but is outside the parent zone's authoritative contents.

A delegation is expressed by NS records at the cut. If a named server lies inside the delegated child, a resolver cannot first resolve that server's name through the child—it needs its address in order to reach the child. The parent therefore supplies an address record called **glue**. Glue is navigation data, not an assertion that the parent is authoritative for every fact about the host.

The root zone delegates top-level domains. A cold recursive resolver starts with a small configured set of root server addresses, asks the root where to find a top-level domain, asks that domain where to find the next child, and continues.

2.3 Wildcards are synthesis rules

A wildcard such as `*.example.com.` does not mean “return this record for every name ending in `example.com.`” It participates only when the queried name does not exist, and the closest-encloser rules determine which wildcard, if any, can synthesize an answer. Existing intermediate names can block a wildcard.

rgbdns stores wildcard records under their literal wildcard owner and its zone lookup searches from the queried name toward the closest existing ancestor. It synthesizes the queried owner in returned records. This is more precise than a string suffix match and is one reason the Zone abstraction tracks known nodes in addition to records.

3 Resource records: typed facts with lifetimes

3.1 The common envelope

Every resource record has:

- an owner name;
- a numeric type;
- a class, almost always Internet class `IN`;
- a time to live, or TTL;
- type-specific data called RDATA.

The TTL is a lease offered to caches. If an authoritative server returns a TTL of 300 seconds, a cache may reuse that answer for at most five minutes before refreshing it. The TTL does not schedule a change and does not guarantee that every cache holds the answer for the full interval. It establishes an upper bound.

Changing a record and then lowering its TTL is too late for clients that already cached the older, longer lease. Planned migrations lower the TTL at least one old-TTL interval before the change, wait, make the change, and later raise it.

3.2 Core types

A maps an owner to an IPv4 address. **AAAA** maps it to an IPv6 address. Several records at one owner form an address RRset; DNS does not promise that clients use them in listed order.

NS names an authoritative server for a zone or delegation.

SOA, the start of authority, identifies the zone and carries operational parameters: primary server, responsible mailbox, serial, refresh, retry, expire, and negative-cache values. A secondary compares serial numbers to decide whether a transfer is needed. Serial arithmetic wraps in a defined 32-bit space, so blindly treating it as an ordinary integer can be wrong near the boundary.

CNAME says that its owner is an alias of another name. Except for DNSSEC and narrowly specified metadata, an owner with CNAME should not also hold unrelated data. A resolver follows the chain while defending against loops and excessive depth.

MX names a mail exchanger and gives it a preference. Lower numbers are preferred. The target is a name, not an address.

PTR provides a name-valued reverse mapping. IPv4 reverse names live below `in-addr.arpa.` with octets reversed. IPv6 reverse names live below `ip6.arpa.` with hexadecimal nibbles reversed.

TXT carries one or more length-delimited byte strings. Presentation formats often make it look like one free-form string, but the wire format retains segments.

SRV names a service endpoint with priority, weight, port, and target.

CAA constrains which certification authorities may issue certificates for a domain.

OPT is not ordinary zone data. It is a pseudo-record used by EDNS to negotiate UDP payload size and carry extension flags and options.

`rgbdns` models these forms with `RecordType`, `Record`, and the `RData` enum in `src/packet.rs`. Known structured types receive structured variants. Unknown types can remain opaque where the format permits, preserving extensibility without confusing untrusted lengths with trusted objects.

3.3 Additional data is an optimization

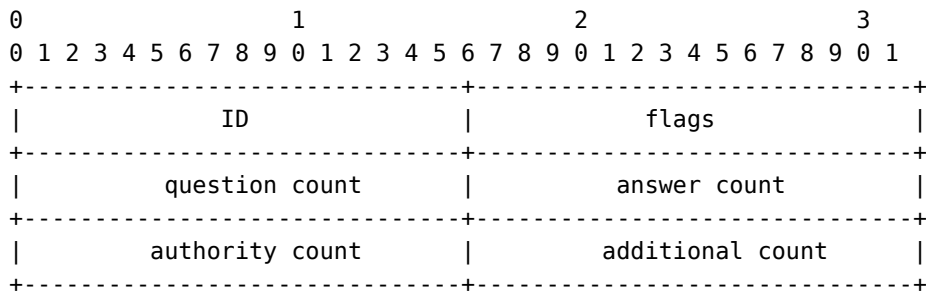
If an answer contains MX, NS, or SRV targets, the server may include associated A and AAAA records in the additional section. This can save queries. It does not change which RRset directly answers the question, and a resolver must apply the correct credibility rules rather than trusting unrelated additional data.

The `rgbdns` authoritative response path collects target names from those record types and adds locally available addresses. It de-duplicates targets before lookup and preserves the distinction between answers and helpful additional.

4 Messages on the wire

4.1 The twelve-byte header

A DNS message begins with a fixed twelve-byte header:



The transaction ID lets a client associate a response with a query. Important flags include QR (query versus response), opcode, AA (authoritative answer), TC (truncated), RD (recursion desired), RA (recursion available), and the four-bit response code.

The four following sections contain questions, answers, authority records, and additional records. A normal question carries a name, requested type, and class. Resource-record sections add TTL, RDATA length, and RDATA.

All multibyte integers are network byte order. Every count and length comes from an untrusted peer. A decoder must prove that bytes exist before reading them, cap allocations, reject invalid labels and pointers, and finish with a coherent message rather than a partially trusted structure.

4.2 Name compression

Repeating full names would waste scarce datagram space. DNS permits a name suffix to be replaced by a two-byte pointer whose high bits are 11 and whose remaining bits are an offset earlier in the message.

Compression turns name decoding into graph traversal. A malicious packet can contain a pointer loop, excessive indirection, or an offset outside the packet. A safe decoder tracks visited offsets or imposes a strict jump bound, checks every target, and separately enforces the expanded 255-octet name limit.

rgbdns's Reader in `src/packet.rs` keeps all reads within a borrowed byte slice. Name decoding validates pointer targets and bounds traversal. Record decoding confines each RDATA parser to the declared RDLENGTH. EDNS option iteration likewise checks the option header and value before advancing.

The Writer performs the reverse operation. Encoding is fallible: counts must fit 16 bits, names and RDATA must fit their fields, and the result must remain valid. This symmetry—decode into typed data, manipulate typed data, encode with checks—is the packet layer's central safety property.

4.3 Errors are protocol results

Several results that sound similar are materially different:

- **NOERROR with answers:** the requested RRset exists.
- **NOERROR without answers**, often called **NODATA**: the name exists but the requested type does not.
- **NXDOMAIN**: the queried name does not exist.
- **SERVFAIL**: the server could not safely complete processing.
- **REFUSED**: policy forbids the operation.
- **FORMERR**: the message is malformed.
- **NOTIMP**: the requested opcode is unsupported.

Negative answers normally include the zone's SOA so resolvers can cache the negative result. Confusing NODATA with NXDOMAIN can suppress other valid types at the same name.

`rgbdns` expresses authoritative lookup outcomes as `Lookup::Answer`, `Referral`, `NoData`, `NxDomain`, and `Refused`. That internal sum type forces the response builder to handle each protocol meaning explicitly.

5 UDP, TCP, EDNS, and truncation

5.1 Why there are two transports

Classic DNS uses UDP for ordinary queries because one request and one response need no connection setup. Traditional UDP DNS assumed a 512-byte message. Larger answers set `TC`, telling the client to retry over TCP. TCP frames every DNS message with a two-byte length.

Zone transfers use TCP. Modern responses—especially DNSSEC responses—often need more than 512 bytes, so EDNS lets a client advertise a larger UDP receive size through an `OPT` pseudo-record. Internet paths can still drop fragmented UDP packets. A commonly conservative payload is 1232 bytes, large enough for useful DNSSEC answers while fitting the IPv6 minimum MTU without fragmentation under normal headers.

TCP is not merely an emergency protocol. Firewalls that assume DNS is UDP-only break standards-compliant resolution.

5.2 Truncation must preserve a valid message

Cutting the last bytes off an encoded message creates a malformed packet. Correct truncation removes complete records, sets `TC`, updates section counts, and re-encodes. A useful removal order discards nonessential additional data before authority and answer data. `OPT` sometimes needs special treatment because it carries the EDNS response.

`src/server.rs` calculates a response limit from the caller's transport limit and the client's EDNS advertisement. It caps advertised UDP size, rejects multiple `OPT` records, responds to unsupported EDNS versions, and constructs a full typed response. If encoding exceeds the limit, `truncate` sets `TC` and removes complete records in a defined order until the packet fits. The same core response logic serves UDP and TCP without treating TCP as a giant UDP datagram.

6 Authority: answering from owned facts

6.1 Finding the closest relevant authority

For a question (name, type), an authoritative server determines:

1. whether the name lies in a served zone;
2. whether a delegation cut is closer than that zone's apex;
3. whether the exact name exists;
4. whether the requested RRset exists;
5. whether a CNAME or wildcard changes the answer;
6. which SOA proves a negative result.

A query beneath a delegated child should produce a referral, not an authoritative negative answer from the parent. A query outside all configured zones should normally be refused. These boundary checks matter more than a simple map lookup.

rgbdns's Zone stores records in a `BTreeMap<Name, Vec<Record>>`, authoritative apices and delegation owners in ordered sets, and separate metadata for location and activation. Lookup walks name ancestry, recognizes cuts, filters visible records, applies exact-name and wildcard rules, and returns the typed Lookup outcome.

The response builder then:

- copies the query ID and relevant RD bit;
- marks authoritative answers with AA;
- expands CNAME chains with a 16-hop limit and visited-name set;
- adds address records for NS, MX, and SRV targets;
- clears AA on referrals;
- adds the SOA to negative answers;
- maps malformed, unsupported, and policy cases to protocol response codes.

The finite CNAME bound and visited set are deliberate denial-of-service and correctness controls. A cyclic zone must not turn one datagram into unbounded work.

6.2 tinydns data as a source language

djbdns uses a compact line-oriented zone source called data. The first character selects a record form. Common forms include:

Prefix	Meaning
.	zone authority plus NS and address data
&	delegation NS and optional glue
Z	explicit SOA
=	A plus matching reverse PTR
+	A only
6	AAAA plus reverse PTR forms
3	AAAA only
@	MX and optional exchanger address
C	CNAME
^	PTR
'	TXT
S	SRV
:	generic record
%	client-location mapping

Fields are colon-separated with octal escapes for bytes that would otherwise be ambiguous. Optional fields carry TTL, timestamp, and location information. The format is terse because it was designed for mechanical generation as well as hand editing.

`Zone::parse` reads this language line by line. It ignores blank, comment, and disabled lines; reports the failing line number; expands convenience forms into ordinary typed records; validates IPv4, flat 32-digit IPv6, names, numeric ranges, and escaped bytes; and records authoritative and delegation structure. When an SOA serial is omitted, file loading derives a nonzero default from the source modification time.

Timestamp fields use TAI64-style cutoffs. Depending on the marker, a record can be visible before or after a specified instant. Location codes select records using configured client IPv4 prefixes. `rgbdns` carries that metadata beside the record and evaluates it at lookup time.

6.3 CDB: compile once, read predictably

The traditional `tinydns-data` compiles text into a constant database, CDB. The serving process reads the compiled file instead of reparsing editable text for every startup or query. Compilation also enables atomic replacement: write and validate a new file, then rename it into place.

`rgbdns`'s `src/cdb.rs` preserves the `djbdns` key/value layout. `compile` serializes typed records and metadata; `load` reads entries and reconstructs a `Zone`. The loader does not trust the database merely because it is local. It bounds file and entry sizes, validates keys, checks record layouts, decodes names and RDATA through explicit lengths, and rejects malformed data.

This is an important general rule: compiled configuration is still input. It may be truncated by a failed copy, generated by an older tool, or replaced by an attacker with filesystem access. Memory safety should not depend on the provenance story being perfect.

7 Recursion: discovering an answer

7.1 Iteration from the root

A recursive resolver turns one client request into a bounded sequence of queries. For `www.example.com`. A, a cold lookup is approximately:

```
stub → recursive resolver
      │ root:      who serves com?
      │ com server: who serves example.com?
      └─ example:   what is www.example.com A?
      ← final answer
```

The resolver follows referrals, resolves nameserver addresses when glue is insufficient, handles aliases, retries servers and transports, and detects loops. It caches useful RRsets so later clients may skip most of this path.

Root hints are not answers to every name. They are bootstrap addresses for reaching the root authority. They need periodic maintenance because the set can change, though names and anycast make changes infrequent.

7.2 The cache is part of correctness

A cache key includes at least name, type, and class. A cached positive RRset expires according to TTL. Negative results also have bounded lifetimes derived from SOA data. Delegation and nameserver-address caches help the iterative algorithm navigate efficiently.

Capacity is as important as time. An attacker can generate endless distinct names. An unbounded cache converts traffic into memory exhaustion. A practical resolver bounds response cache bytes, nameserver cache entries, recursion depth, referral work, packet sizes, concurrent operations, and timeouts.

`src/bin/dnscache.rs` uses Hickory's recursive zone handler inside `rgbdns`'s process and policy shell. It configures:

- randomized query-name letter case;
- a bounded response cache, defaulting to 16 MiB;
- a bounded nameserver cache;
- bounded ordinary and nameserver recursion depth;
- a 1232-byte EDNS payload;
- UDP and TCP listeners;
- loopback-only clients by default, expanded through `ALLOW_NETS`.

Configuration values are parsed with explicit minimums and maximums. A typo such as an enormous cache size fails startup rather than silently allocating an operator's mistake.

7.3 Forward zones and djbdns roots

Private namespaces and split DNS often need selected suffixes sent to specific servers. `rgbdns` reads forward-zone configuration from the environment and the `djbdns`-style `R00T/servers` directory. The filename identifies a suffix and the file lists bounded server addresses.

The special `servers/@` file represents root servers. Hickory consumes a root hints file in master-file syntax, so `PreparedRoots` translates `djbdns`'s plain address list into a private temporary file. Creation uses restrictive permissions and cleanup occurs when the prepared object is dropped. This adapter preserves the external configuration contract without weakening the library boundary.

Forwarded private zones disable strict case-randomization response matching because legacy authorities may canonicalize owner case. They retain TCP retry and a bounded cache. This is a scoped compatibility decision, not a global removal of query hardening.

8 DNSSEC: authenticating the chain

8.1 What ordinary DNS cannot prove

Transaction IDs, source ports, and query-case randomization make blind spoofing harder, but they do not cryptographically establish who published an RRset. DNSSEC adds signatures and a chain of trust.

A zone signs RRsets with private keys and publishes DNSKEY records. A parent publishes a DS digest that identifies a child key. Starting from a configured root trust anchor, a validating resolver can authenticate the root DNSKEY, then a top-level domain's DS and DNSKEY, and so on to the answer.

RRSIG authenticates an RRset over a validity interval. DS links parent to child. NSEC or NSEC3 authenticates nonexistence by proving gaps in the ordered namespace. DNSSEC provides origin authentication and integrity; it does not encrypt queries or hide names.

Validation outcomes matter:

- **secure**: a valid chain reaches the answer;
- **insecure**: the chain proves that the child is unsigned;
- **bogus**: signatures or proofs fail;
- **indeterminate**: validation cannot be completed safely.

A resolver must not turn bogus data into a normal answer merely to improve availability. Clock correctness also becomes a dependency because signatures have inception and expiration times.

8.2 `rgbdns` validation policy

`rgbdns` configures the recursive handler with a static root trust anchor and DNSSEC validation enabled. Validation and NSEC3 work receive bounded caches and iteration policies. A failed validation surfaces as resolution failure rather than an unchecked answer.

The authoritative `rgbdns` data path focuses on the `djbdns` record surface; the recursive path is where DNSSEC validation is currently integrated. This is an example of honest component boundaries: “the suite supports validating recursion” does not imply that every authoritative signing workflow has been recreated.

9 Zone transfer and secondary service

9.1 AXFR is a stream, not a giant datagram

AXFR transfers a complete zone over TCP. A successful stream begins with the zone’s SOA, contains the zone records, and ends with the SOA again. The records may span many DNS messages. A client must continue until it sees the closing SOA under the transfer rules; reading one response is insufficient.

Transfers reveal the zone contents and can consume resources, so authorities normally restrict clients. TSIG is a common authentication mechanism in the wider ecosystem, while IP allowlists are a simpler policy with weaker identity properties.

`src/axfr.rs` provides both sides. `axfrdns` accepts TCP only and checks client networks, loopback by default. It requires one AXFR question, obtains a boundary-aware transfer from Zone, and frames bounded messages. `Zone::transfer` excludes records beneath delegated child zones and wraps the result in the apex SOA.

`axfr-get` generates a random transaction ID, validates response identity and shape, collects records until the closing SOA, renders them in `tinydns` source form, writes a temporary output, and atomically installs the completed file. The temporary/final path pair prevents a failed transfer from replacing usable data with a partial zone.

10 The `rgbdns` program family

10.1 One suite, small purposes

`rgbdns` deliberately exposes separate commands:

Command family			Purpose
tinydns,	tinydns-data, tinydns-edit	tinydns-get,	authoritative service and data maintenance
	dnscache		validating recursive resolver and cache
	axfrdns, axfr-get		zone transfer server and client
	rbldns, rbldns-data		address-prefix blocklist DNS
	pickdns, pickdns-data		location-aware address selection
	walldns		synthetic address/reverse answers
dnsq, dnsqr, dnsip*,	dnsname, dnsmx, dnstxt		queries and diagnostics
	dnsfilter, dnstrace, random-ip		stream lookup, delegation tracing, testing
	*-conf		service-directory generation
setuidgid, multilog,	tai64n, tai64nlocal		process and logging support

This composition makes privilege and failure boundaries visible. A compiler can run with write access to data while the server runs read-only. A recursive cache can be restarted without touching authority. Diagnostic clients reuse the packet and client libraries rather than embedding daemon behavior.

10.2 Specialized responders

rbldns treats the labels before a configured suffix as a numeric address, finds the most-specific matching IPv4 prefix in a compiled database, and returns configured A/TXT data. Parsing caps the number of numeric labels and validates networks before compilation.

pickdns maps client prefixes to two-byte locations and selects address sets for that location. It shuffles eligible addresses with operating-system randomness. Location-aware answers are a policy feature; clients behind shared resolvers may appear at the resolver's address, a limitation operators must understand.

walldns synthesizes narrowly defined forward and reverse answers without a zone database. These specialized services run through `src/special.rs`, which provides shared bounded UDP/TCP serving and passes the peer address to the handler.

The lesson is architectural: once parsing, transport, names, and record models are sound, unusual DNS policies can be small pure response functions rather than new monolithic servers.

11 Client behavior and diagnostics

11.1 A query is more than sending bytes

A DNS client creates a random transaction ID, encodes one question, sends it to an intended server, receives a response, and validates at least:

- source endpoint where the transport permits;

- transaction ID;
- QR and response shape;
- matching question;
- declared section lengths and names;
- truncation, with TCP retry when needed.

`src/client.rs` reads `DNSCACHEIP` or `/etc/resolv.conf`, supports IPv4 and IPv6 socket syntax, gets IDs from the operating system, applies UDP timeouts, rejects mismatched responses, and retries truncated UDP replies over TCP. The small command binaries format results for different use cases, while `dnsq` allows an explicit server and `dnsqr` uses recursive configuration.

`dnstrace` is conceptually different from a recursive lookup: it exposes the delegation path and intermediate authority so an operator can see where the chain stops. Good diagnosis asks four separate questions:

1. What did the stub send?
2. What did the recursive resolver cache or validate?
3. What delegation did the parent publish?
4. What does the authoritative server say directly?

Testing only the final application collapses all four layers and encourages guessing.

11.2 Practical checks

When a name fails, inspect type, server, flags, and authority section rather than asking only whether an address appeared.

```
dnsq A www.example.com 192.0.2.53
dnsq AAAA www.example.com 192.0.2.53
dnsq SOA example.com 192.0.2.53
dnstrace A www.example.com
```

Compare UDP and TCP when answers are large. Query the parent-side NS records and the child authority separately. An NXDOMAIN with an SOA is different from a timeout, SERVFAIL, or REFUSED, and each points to a different layer.

12 Security engineering in `rgbdns`

12.1 The packet is hostile

DNS combines nearly every parser hazard: nested lengths, compression pointers, variable counts, binary strings, recursive relationships, and network-facing availability requirements. “Written in Rust” removes broad classes of memory corruption, but it does not automatically prevent allocation bombs, infinite loops, CPU amplification, path races, policy errors, or accepting incoherent messages.

`rgbdns` therefore uses several layers:

- `#![forbid(unsafe_code)]` for the library;
- explicit bounds before every wire read;

- validated Name, Message, and RData objects;
- limits on compression traversal, aliases, records, files, configuration lists, recursion, transfers, and cache sizes;
- cryptographic operating-system randomness for query IDs and selection;
- complete-record truncation;
- loopback-only defaults for recursion and transfer;
- atomic replacement for compiled databases and fetched zones;
- no shell interpolation when replacing a process.

Property tests in `tests/packet_properties.rs` feed arbitrary bytes to the decoder and exercise encode/decode invariants. Golden CDB fixtures compare compiled output with the expected djbdns layout. Network tests cross real UDP and TCP boundaries. Compatibility tests are valuable here because a parser can be safe yet subtly wrong, or compatible yet unsafe.

12.2 Least privilege and filesystem boundaries

The `*-conf` commands generate service directories whose run scripts execute the daemon under a selected account. `rgbdns's setuidgid` resolves the user and group, initializes supplementary groups, drops GID and UID, verifies the result, and directly replaces itself with the target program. Direct replacement preserves signals and exit status and avoids an extra shell-owned process.

Generated paths are shell-quoted and support binaries by absolute path. Configuration writers reject unsafe existing file types and apply intentional modes. CDB and AXFR update workflows install only complete outputs.

Privilege dropping is not a substitute for a restricted service account, read-only data, network policy, or supervisor hardening. It is one layer in a deployment.

13 Time and logs: TAI64N

13.1 Why a DNS suite contains time tools

Long-running daemons need logs, and djb's tools use TAI64N labels. A label has an @, sixteen hexadecimal digits of biased TAI seconds, and eight hexadecimal digits of nanoseconds:

```
@4000000037c219bf2ef02e94
```

TAI is a continuous atomic timescale. UTC inserts leap seconds, so converting between a POSIX/UTC timestamp and TAI requires the applicable TAI-UTC offset. The offset was 10 seconds at the Unix epoch convention used here and reached 37 seconds after the 2016 leap second.

`tai64n` timestamps each input line at the moment its first bytes are read. `tai64nlocal` recognizes a valid leading label and replaces it with local civil time at nanosecond precision. Invalid prefixes pass through unchanged. `multilog -t` uses the same label generator, ensuring standalone filters and log files agree.

`src/tai64.rs` contains the complete 1972–2017 positive-leap transition table. It validates the fixed-width hexadecimal form and nanosecond range. Both stream filters use bounded memory:

the localizer buffers only the 25-byte candidate prefix and streams the rest of even an extremely long line. At I/O failure the command exits 111, following the daemontools convention.

TAI64N provides sortable, unambiguous event labels. Converting to local time is a presentation step, not the archival representation.

14 Running rbgdns under supervision

14.1 The service contract

rgbdns daemons run in the foreground, emit diagnostics to standard error, take configuration from files and environment, and terminate on fatal startup errors. That is the portable contract a supervisor needs. The generated djbdns-style directories additionally provide `run` and `log/run` programs, but the daemon binaries do not require a particular supervisor.

The classic daemontools control plane is:

```
supervise service/      keep one process running
svc -u service/         bring it up
svc -d service/         bring it down
svc -t service/         send TERM
svstat service/         inspect status
```

No modern replacement is universally best. Choose according to the host and the migration boundary.

14.2 Recommendations

14.2.1 Existing Linux host: systemd

Use systemd when the machine already boots and manages services with systemd. It supplies dependency ordering, restart policy, socket and readiness models, resource controls, credential and filesystem sandboxing, a unified journal, and distribution-native administration. Avoid wrapping an rbgdns daemon in a second nested supervisor.

A minimal authoritative unit is:

```
[Unit]
Description=rgbdns authoritative DNS
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
User=dns
Group=dns
Environment=IP=192.0.2.53
Environment=PORT=53
Environment=DATA=/etc/rgbdns/data.cdb
ExecStart=/usr/local/bin/tinydns
Restart=on-failure
```

```
RestartSec=1s
NoNewPrivileges=yes
ProtectSystem=strict
ProtectHome=yes
PrivateTmp=yes
ReadOnlyPaths=/etc/rgbdns
AmbientCapabilities=CAP_NET_BIND_SERVICE
CapabilityBoundingSet=CAP_NET_BIND_SERVICE
```

```
[Install]
WantedBy=multi-user.target
```

Prefer a socket above 1024 or a narrowly bounded bind capability over running the daemon as root. Test hardening settings on the target distribution because name-service libraries and trust-anchor paths may require additional read-only access.

Command mapping:

daemontools	systemd
<code>svc -u service</code>	<code>systemctl start service</code>
<code>svc -d service</code>	<code>systemctl stop service</code>
<code>svc -t service</code>	<code>systemctl kill --signal=TERM service</code>
<code>svc -h service</code>	<code>systemctl kill --signal=HUP service</code>
<code>svstat service</code>	<code>systemctl status service</code>
multilog output	journal, or explicit file logging policy

14.2.2 Closest service-directory migration: runit

Use runit when you want the smallest conceptual migration from daemontools. It uses a service directory with a run program, keeps the supervised process in the foreground, has a companion log service, and exposes the compact `sv` control command. Existing `rgbdns` generated run scripts are close to the required shape; adjust the directory layout and enablement symlink for the distribution.

daemontools	runit
<code>svc -u service</code>	<code>sv up service</code>
<code>svc -d service</code>	<code>sv down service</code>
<code>svc -t service</code>	<code>sv term service</code>
<code>svstat service</code>	<code>sv status service</code>

Choose runit for minimal hosts, appliances, or migrations where preserving the service-directory model matters more than rich dependency and sandbox policy.

14.2.3 Strong supervision composition: s6 and s6-rc

Use s6 when precise process supervision, reliable readiness, and composable small tools are primary requirements. Its s6-supervise and s6-svc are close in spirit to supervise and svc; s6-rc adds declared dependencies and transactional service-state changes. The ecosystem is particularly effective in carefully constructed containers and small systems, but its compilation and directory conventions make migration more involved than runit.

daemontools	s6
supervise service	s6-supervise service
svc -u service	s6-svc -u service
svc -d service	s6-svc -d service
svc -t service	s6-svc -t service
svstat service	s6-svstat service

Choose s6/s6-rc when the team is willing to own its service database and wants more rigorous dependency transitions than ad hoc shell orchestration.

14.2.4 OpenRC and container orchestrators

On an OpenRC-based distribution, use the native init integration unless there is a deliberate reason to introduce another supervision tree. OpenRC service scripts can use its supervisor support while retaining distribution-standard boot ordering and administration.

In Kubernetes or a similar orchestrator, run one foreground rgbdns daemon per container and let the platform own restart, health, resource limits, log collection, and rollout. Use a Deployment for tinydns or dnscache, a Service for stable network reachability, readiness/liveness probes that test the intended DNS role, ConfigMaps or mounted immutable CDBs for public data, and Secrets for sensitive material. Do not put systemd, daemontools, and the orchestrator around the same single process.

An s6-based container is reasonable only when one image intentionally contains several cooperating long-lived processes and that tradeoff is explicit.

14.3 A practical selection rule

Use this order:

1. Follow the host's native manager: systemd on systemd hosts, OpenRC on OpenRC hosts.
2. For a direct service-directory replacement, choose runit.
3. For a designed supervision graph or multi-process container, choose s6/s6-rc.
4. In an orchestrated single-process container, use the orchestrator.

The least risky migration preserves one owner for restart policy and logs. Running two supervisors creates ambiguous signal paths, duplicate restarts, and status commands that disagree.

15 Operating an authoritative service

15.1 Build, stage, verify, replace

A safe publication cycle separates source editing from serving:

```
cd /etc/rgbdns
tinydns-data
tinydns-get example.com A www.example.com
```

In production, compile in a staging directory, run representative exact, wildcard, delegation, negative, IPv4, IPv6, and large-response queries, then atomically replace `data.cdb`. Retain the previous known-good database for rollback. Query the bound service over both UDP and TCP after deployment.

Do not expose the recursive service to arbitrary networks by accident. The default `ALLOW_NETS` is loopback only because an open resolver can be abused for amplification and can consume local capacity. Likewise, expand AXFR allowlists only for intended secondaries.

15.2 Observe the right signals

Useful signals include:

- query and error rate by transport;
- truncated UDP responses and TCP retries;
- SERVFAIL, REFUSED, NXDOMAIN, and validation-failure rates;
- resolver cache capacity and latency percentiles;
- process restarts and file-descriptor use;
- root-hint and trust-anchor freshness;
- time synchronization;
- CDB build identity and deployment time.

High NXDOMAIN volume is not automatically an incident; browsers, typo traffic, and discovery protocols generate it. A change from baseline paired with latency or SERVFAIL is more meaningful.

TAI64N log labels make events stable for storage. Convert them for human display at the edge:

```
tail -f main/current | tai64nlocal
```

16 Testing DNS software

16.1 Layers of evidence

Unit tests establish local invariants: name limits, record parsing, lookup outcomes, leap conversion. Property tests explore parser state spaces that examples miss. Golden fixtures establish compatibility with an external file format. Integration tests cross process and socket boundaries. Live interoperability tests compare behavior with independent clients and servers.

rgbdns uses all of these. A useful local sequence is:

```
cargo fmt --check
cargo test
cargo clippy --all-targets --all-features -- -D warnings
```

Network tests bind unprivileged loopback ports. CDB tests compile canonical fixtures and compare entries. Daemontools tests exercise process replacement, rotation, and TAI64 filter behavior. Packet properties assert that arbitrary input does not panic and that supported structured messages survive encode/decode round trips.

16.2 Adversarial cases worth keeping

Every DNS implementation should retain regression cases for:

- a compression pointer to itself or a pointer cycle;
- a pointer or RDATA length just beyond the packet;
- maximum-length labels and names;
- counts that cannot be satisfied by the remaining bytes;
- duplicate or malformed OPT records;
- tiny advertised transport limits;
- CNAME loops and excessive chains;
- wildcard names blocked by existing nodes;
- delegation cuts beneath an authoritative apex;
- NODATA versus NXDOMAIN;
- AXFR without a closing SOA;
- an enormous log line;
- configuration counts at and beyond each bound.

Tests should assert protocol meaning, not only that the process remains alive. A safe FORMERR is better than a crash, but a silent NOERROR can still be a serious bug.

16.3 Conformance as an executable specification

The conformance suite turns protocol prose into named, reviewable cases. Its scope is the DNS surface rbgdns implements; it does not imply support for every extension ever assigned by IANA. The principal coverage is:

Standard	Behavior exercised
RFC 1035	header identity, flags, names, compression, typed RDATA, UDP and TCP results
RFC 2181	in-bailiwick glue, coherent RRset TTLs, duplicate suppression, CNAME exclusivity
RFC 2308	NXDOMAIN versus NODATA, authoritative SOA, negative-cache TTL
RFC 3597	unknown QTYPE behavior and lossless opaque RDATA
RFC 4343	case-insensitive name identity with query-case preservation
RFC 4592	closest-encloser wildcard synthesis and empty non-terminals
RFC 5936	AXFR framing, identity, flags, SOA bookends, and zone boundaries
RFC 6891	one root-owned OPT, payload negotiation, DO, BADVERS, and unknown options
RFC 7766	TCP framing, connection reuse, pipelining, and full-size responses
RFC 8906	the authoritative-server matrix for unknown types, opcodes, flags, and EDNS fields
RFC 9619	exactly one question in a standard query

This is more useful than a single “RFC compliant” label. A test name identifies the rule, a packet fixture demonstrates it, and a failure points to a specific semantic regression.

RFC 8906 is especially valuable because it tests how a server behaves at the edges of what it understands. An unknown ordinary type is not a protocol error: the answer depends on whether the owner name exists. An unknown opcode is different. Because that opcode may define a body layout unlike QUERY, the server must produce NOTIMP from the header without first interpreting the body as an ordinary question. Unknown EDNS options are structurally validated and then ignored. An unsupported EDNS version produces BADVERS while retaining an OPT response.

The independent `drill` integration test supplies another boundary. It launches the real `tinydns` binary and asks the `ldns` client to make UDP, TCP, EDNS, mixed-case, and unknown-type queries. This catches accidental agreement between `rgb dns`’s own encoder and decoder: the request and response cross an implementation boundary.

The complete focused matrix is:

```
cargo test --test rfc_conformance
cargo test --test wire_security
cargo test --test packet_properties
cargo test --test drill_interop
```

The generated suite exercises forty thousand cases per complete run. It feeds arbitrary bytes to the decoder, reparses every accepted packet, generates structured messages for semantic round trips, and changes ASCII letter case without changing DNS name identity. A separate truncation corpus tries every prefix of a valid structured packet. These properties do not prove the absence of all parser defects, but they explore combinations that hand-written examples rarely anticipate.

16.4 Hardening found by conformance work

Conformance testing improved the implementation rather than merely describing it.

The name decoder now records valid prior name boundaries. A compression pointer must be backward *and* must target one of those boundaries. Merely pointing at earlier bytes that happen to resemble a label sequence is rejected. This closes a class of ambiguous parses without forbidding legal compression.

Stub responses are bound to the request ID, QR bit, opcode, and exact question. TCP responses carrying TC are rejected. AXFR applies the same identity checks and additionally requires authoritative, non-truncated messages, controlled question repetition, an empty authority section, matching opening and closing SOAs, and records confined to the requested zone. These rules prevent a plausible-looking but unrelated response from being accepted as the answer to the outstanding operation.

Zone loading rejects a CNAME owner that also has other data and rejects multiple different CNAME targets. Before transmission, RRsets are normalized to their minimum TTL and duplicate records are removed. Negative answers cap the SOA TTL at the SOA MINIMUM field as RFC 2308 requires. EDNS OPT records in the wrong section and duplicate OPT records produce FORMERR.

The UDP and TCP daemons now share one bounded transport module. TCP connections carry deadlines, use a fixed worker pool, accept multiple framed queries, and support pipelined requests. This removes duplicated socket code while making RFC 7766 behavior an invariant shared by the authoritative and specialized servers.

16.5 Benchmarks and evidence-driven optimization

Correctness gates run before performance conclusions. The benchmark is a dependency-free stable-Rust harness in `benches/dns_core.rs`; the same harness is available as `examples/dns_core_bench.rs` for quick release-mode runs:

```
cargo bench --bench dns_core
RGBDNS_BENCH_ITERATIONS=10000 \
  cargo run --release --example dns_core_bench
```

It warms every operation, passes values through `std::hint::black_box`, and reports nanoseconds per operation. Measurements are comparable only on the same host, toolchain, power state,

and iteration count. Wire size is reported beside CPU time because DNS compression exchanges encoder work for fewer network bytes.

The July 2026 checkpoint used release mode on one aarch64 Android host:

Operation		Baseline	Optimized	Result
Encoded	64-record response	2,147 bytes	1,059 bytes	50.7% smaller
Decode	small query	542 ns	458 ns	15.5% faster
Decode	64-record response	52,661 ns	29,540 ns	43.9% faster
Encode	64-record response	2,318 ns	5,309 ns	2.3 times slower
Exact	lookup, 1,000 names	1,262 ns	1,244 ns	1.4% faster
	NXDOMAIN, 1,000 names	29,889 ns	2,726 ns	11.0 times faster
Small	authoritative response	17,007 ns	7,714 ns	54.6% faster
Truncate	200-record response	3,098,232 ns	2,570,077 ns	17.0% faster

Three structural changes explain most of the gains.

First, Zone maintains an index of every node, including empty non-terminals. A clearly absent name can return NXDOMAIN without scanning the records of a thousand-name zone. Conditional records still take the visibility path, so the index does not erase time or location semantics.

Second, response truncation searches the number of tail records to remove instead of encoding once for every removed record. It preserves the question and OPT record as long as possible and validates the final candidate against the transport limit.

Third, the packet writer records complete names and suffixes for RFC 1035 compression. RRsets tend to repeat the immediately preceding owner, so a last-owner cache avoids rebuilding and hashing suffix keys on the dominant path. The first compression design encoded the 64-record case in 34,075 ns; the cache reduced that to 5,309 ns.

The remaining encoder regression is intentional and visible. Compression makes the example packet roughly half as large while taking more local CPU than the old uncompressed writer. That is a defensible trade for an authoritative server because it reduces datagram pressure, TCP bytes, and downstream decode work. Recording the regression matters: optimization should reveal tradeoffs, not hide them behind one favorable number.

17 Reading the rbgdns source

17.1 A path through the code

Read the project in dependency order:

1. `src/name.rs` — the foundational name invariant.
2. `src/packet.rs` — types and bounded wire codec.
3. `src/zone.rs` — tinydns source and authoritative lookup semantics.
4. `src/cdb.rs` — compiled compatibility format.
5. `src/server.rs` — query validation, answer construction, transport limits.
6. `src/client.rs` — stub behavior and TCP fallback.
7. `src/axfr.rs` — streaming zone movement and atomic installation.
8. `src/dnscache_config.rs` and `src/bin/dnscache.rs` — iterative recursion, DNSSEC, forwarding, access, and resource policy.
9. `src/rbl.rs`, `src/pick.rs`, `src/wall.rs`, and `src/special.rs` — specialized responders.
10. `src/conf.rs`, `src/setuidgid.rs`, `src/multilog.rs`, and `src/tai64.rs` — deployment and operations.

The binaries in `src/bin` should then look thin. That is intentional. They parse the command contract, load configuration, call a library boundary, print diagnostics, and map fatal errors to the suite’s exit convention.

17.2 Design patterns to carry elsewhere

Several rbgdns choices generalize beyond DNS.

Parse into valid types. If an invalid name can circulate as an ordinary string, every consumer must rediscover validation.

Bound dimensions independently. A packet byte limit does not replace a compression-depth limit; a cache byte limit does not replace a recursion-depth limit.

Separate policy from mechanism. `transport.rs` owns bounded UDP and TCP mechanics while the authoritative and specialized handlers own answer policy.

Compile mutable source into immutable serving data. This gives validation, atomic rollout, simple readers, and easy rollback.

Preserve protocol distinctions internally. A `Lookup` enum prevents `NXDOMAIN`, `NODATA`, referral, and refusal from collapsing into “no records.”

Run in the foreground. It composes with old and new supervisors and keeps signals understandable.

Treat compatibility files as hostile. Historical layout fidelity need not mean historical trust assumptions.

18 Where DNS ends

DNS establishes named, cacheable facts and, with DNSSEC, their authenticated origin. It does not prove that the address belongs to the application a user intended, encrypt the subsequent connection, guarantee freshness inside the TTL window, or choose a healthy endpoint. TLS identity, application discovery, load balancing, routing, and monitoring build on DNS but remain separate systems.

That boundary is the best final replacement for the phone-book metaphor. DNS is a delegated publication and discovery protocol. Its tree assigns authority; its records carry typed statements; its TTLs make caching explicit; its packet format makes efficient exchange possible; recursion joins many authorities into one answer; DNSSEC authenticates the chain; and supervision keeps the implementing processes available without becoming part of the protocol.

rgbdns expresses those ideas as small programs over shared, validated Rust types. Understanding the protocol makes the program family unsurprising. Reading the program family, in turn, shows how the abstract DNS model becomes bounded packets, immutable databases, iterative queries, atomic files, and foreground processes.

19 Appendix A: Configuration quick reference

Common daemon variables include:

Variable	Meaning
IP	listen address
PORT	listen port
DATA	authoritative text or CDB path where supported
ALLOW_NETS	comma-separated client CIDRs for recursion or transfer
DNSCACHEIP	recursive endpoints used by client tools
CACHESIZE	bounded recursive response-cache capacity
NSCACHESIZE	bounded nameserver-cache entries
RECURSION_LIMIT	ordinary recursion depth
NS_RECURSION_LIMIT	nameserver-resolution recursion depth
ROOT	djb dns-compatible resolver configuration root

Use the command's *-conf generator as a starting point, then adapt the foreground run contract to the chosen native supervisor.

20 Appendix B: Further reading

The protocol is defined across many RFCs. A productive sequence is:

- RFC 1034, *Domain Names—Concepts and Facilities*.
- RFC 1035, *Domain Names—Implementation and Specification*.
- RFC 2181, clarifications including RRset and credibility rules.
- RFC 2308, negative caching.
- RFC 6891, Extension Mechanisms for DNS (EDNS(0)).
- RFC 7766, DNS over TCP requirements.
- RFC 4033, RFC 4034, and RFC 4035, DNSSEC.
- RFC 5155, NSEC3.
- RFC 5936, AXFR.
- RFC 1982, serial number arithmetic.

Implementation and operational references:

- The djbdns documentation: <https://cr.yp.to/djbdns.html>
- TAI64N format and tools: <https://cr.yp.to/daemontools/tai64n.html>
- s6 overview: <https://skarnet.org/software/s6/overview.html>
- s6-rc overview: <https://skarnet.org/software/s6-rc/overview.html>
- runit benefits: <https://smarden.org/runit/benefits.html>
- systemd project documentation: <https://systemd.io/>
- Hickory DNS: <https://hickory-dns.org/>